

EPRINC Oil and Gas Cushion Index

Methodology, version 1.6

Draft

1. Purpose

The Cushion Index measures the physical buffer available in the United States and global oil and natural gas system against an unexpected supply loss.

It answers one question. How much slack does the system have. It is not a price index, it does not forecast, and it makes no claim about electricity reserves, nuclear availability, coal stocks, or transmission. The narrower scope is deliberate.

The index exists because price alone is a poor indicator of physical security. Buffers can be thin while prices are low, and the resulting fragility is invisible in market data until it is expensive. The Cushion Index is designed to register that condition.

Interpretation. Scores run 0 to 100. Higher indicates more physical slack. A score of 50 is defined as physically adequate cushion, not as the historical median.

2. Scoring approach

Each component converts to a 0 to 100 score using fixed, transparent thresholds anchored to operational conditions. Historical data calibrate and validate those thresholds. They do not determine the score mechanically.

This is a deliberate departure from percentile scoring, which was used in an earlier draft. Percentile scoring made comfortable mean only comfortable relative to 2010 through 2025, which is a weaker claim than this index intends to make. A physical security index should be able to say that conditions are thin in absolute terms.

2.1 Method

For each component, a piecewise linear function maps the underlying metric to a 0 to 100 score through published anchor points, bounded at both ends.

Score equals the minimum of 100 and the maximum of 0 and the interpolated value.

Anchors are published in full in Section 5 and frozen for the version. Values beyond the outer anchors cap at 0 or 100. These caps mark where additional cushion, or additional erosion, stops being differentiating for the security question. They are scoring caps, not claims about the physical minimum or maximum the system could reach.

2.2 Anchor tiers

Not all components are anchored equally, and the document distinguishes them rather than overclaiming.

Operationally anchored thresholds. Outer and critical thresholds are tied where possible to physical or operational constraints, such as tank bottoms, minimum operating inventory, or the point at which spare capacity reaches zero. Intermediate adequacy thresholds reflect documented analytical judgment rather

than a physical law. This applies to spare capacity, all days of supply measures, and the domestic gas margin family.

Fixed mapping onto a seasonal baseline. Gas storage adequacy is the deliberate exception. There is no absolute volume of working gas that is adequate independent of season, so this component applies fixed thresholds to deviation from a rolling five year seasonal baseline. The mapping is fixed and transparent, but the baseline itself is historical. Fifty means normal for that week, not a fixed physical quantity.

2.3 What is not used

Historical percentile ranking is not used to produce any score. It appears only in validation.

Historical min max normalization is explicitly rejected. It resembles absolute bounding without providing it. A single unusual observation can define the scale permanently, and future values can fall outside 0 to 100. The 2020 demand collapse produced such observations across nearly every petroleum series. Where the index uses bounds, they are operational, never historical extremes.

2.4 Re-benchmarking and accepted erosion

Operationally anchored thresholds do not re-benchmark. If refining capacity or storage adequacy erodes structurally over a decade, the affected components will read progressively worse and will not adjust to treat the new level as normal. This is intended. Percentile scoring would conceal secular erosion by silently redefining the baseline. This index is designed to register it.

Gas storage adequacy is the stated exception to this principle, since its fixed scoring function is applied to a rolling seasonal baseline as described in Section 2.2. As of this version, gas storage adequacy carries 25 percent of total index weight, so three quarters of the index retains full sensitivity to secular erosion while one quarter re-benchmarks by design.

3. Structure

Two sub-indices, weighted equally, each published independently.

Sub-index	Weight	Components
Gas cushion	50%	2, at 25% of total each
Oil cushion	50%	3, at 16.7% of total each

Within each sub-index, components are weighted equally. The only discretionary weighting in the entire structure is oil versus gas at 50 to 50.

Divergence between sub-indices is itself reportable. Oil adequate, gas strained carries more information than any single figure.

No component is inverted. Every input runs the same direction. Higher equals more cushion. Refinery utilization, which was inverted in an earlier draft, has been removed from scoring for the reason described in Section 4.5.

4. Components

4.1 Gas storage adequacy, 25 percent of total

Measures whether working gas is adequate and located where it will be needed.

Calculation. For each of five EIA storage regions, compute deviation from that region's own same week five year average as a percentage. Weight the five regional deviations by regional January gas consumption share. Apply the Section 5.1 threshold function to the weighted composite.

Regional weighting is included from the outset rather than deferred, since national aggregation would treat Gulf Coast salt dome inventory as interchangeable with New England supply, which it is not.

Regional weights are derived from state level residential plus commercial natural gas consumption in the Natural Gas Monthly, aggregated to storage regions, using January average daily consumption share across 2015 through 2025. Residential plus commercial is used rather than total consumption deliberately. Total consumption includes industrial and power burn, which run year round and are heavily concentrated in Texas and Louisiana, and weighting by total would give the South Central region a large share for reasons unrelated to cold weather vulnerability. Residential plus commercial is the weather sensitive portion of demand and is the relevant measure for a cold snap security question. These are called January residential and commercial consumption share weights, not winter peak demand weights, since a monthly average is not a peak. Weights are fixed, recomputed annually, and versioned. Alaska and Hawaii are excluded because the weekly storage series cover the Lower 48 only.

Current weights, calibrated on January 2015 through 2025: East 0.4052, Midwest 0.3008, South Central 0.1118, Mountain 0.0924, Pacific 0.0898. The state to region mapping is verified against EIA's published storage region definitions and reproduced in the appendix. EIA last redefined these regions in October 2015.

Vintage rule. Each historical observation uses the five year average applicable to that week at the time, never the current five year average.

Source. EIA Weekly Natural Gas Storage Report, Thursdays.

Ladder widened after backtest review. The original anchors, spanning plus or minus 25 percent deviation, saturated at 0 or 100 for extended real periods, including 2013 to 2014, 2016, 2020, and 2024, which collapsed the distinction between a routine cold snap and a genuine crisis. The real historical range runs from roughly negative 55 to positive 47. Anchors were widened accordingly and reverified, and only the 2014 polar vortex trough, the worst point in the 2010 through 2025 record, now touches the floor of the ladder.

4.2 Domestically retained supply, 25 percent of total

Measures the share of the domestic gas supply base retained for domestic use rather than committed to export offtake. Storage measures the stock. This measures a structural commitment condition.

Naming. An earlier draft called this uncommitted supply share, which overclaimed. The gas not exported is not uncommitted in any meaningful sense, since most of it is committed to domestic power, industrial, residential and commercial use. What the measure actually captures is how much of the supply base is retained domestically and therefore remains available to a domestic demand shock on short notice, as against volume already contracted to leave the country. The name was changed to reflect that.

Calculation, trailing twelve months.

Uncommitted share equals one minus the sum of LNG feedgas and pipeline exports, divided by the sum of dry gas production and pipeline imports.

This metric replaced an earlier candidate, the domestic gas supply margin, defined as production plus imports minus exports minus feedgas, divided by consumption. That candidate was tested and rejected.

Over a twelve month window the gas balance is close to an accounting identity, so the margin ratio stayed within roughly one percentage point of 1.0 across fifteen years of real data regardless of underlying conditions, and its remaining variation correlated with gas storage adequacy at 0.62, meaning it was not measuring anything the storage component did not already capture. Domestically retained supply is not an identity. It showed real range, roughly 15 percentage points between the tenth and ninetieth percentile of history, and correlated with storage adequacy at only negative 0.24, indicating genuine independence.

Ladder rationale. There is no physical floor or ceiling for this metric, since committed share is a policy and contractual condition rather than an inventory level. Anchors are expressed in terms of committed share, the complement of uncommitted share, and are stated as scoring caps rather than physical bounds.

Committed share	Uncommitted share	Score
35% or more	0.65 or less	0
28%	0.72	25
21%	0.79	50
14%	0.86	75
7% or less	0.93 or more	100

The floor anchor was widened from an earlier draft value of 25 percent committed after review found the growth rate of committed share, roughly 1.1 percentage points per year across 2011 through 2025, would reach a 25 percent floor within 4 to 5 years given the known pipeline of permitted LNG capacity still coming online through the late 2020s. That is well inside this index's working horizon, so the earlier floor would have caused premature saturation almost immediately after launch. The revised floor of 35 percent committed provides working headroom through the currently visible export buildout. Backtest review confirmed committed share climbed from under 5 percent in 2011 through roughly 20 percent by mid 2025, tracking the sequence of Sabine Pass, Cameron, Corpus Christi, and Plaquemines coming online, with no reversals. The Freeport LNG outage in June 2022 did not produce a spurious spike, indicating the known limitation described below is not currently distorting this series.

This ladder should be revisited again once the currently permitted LNG buildout is substantially complete, since the growth rate that motivated this widened floor will not continue indefinitely.

Known limitation. LNG exports are used as a proxy for feedgas, since EIA does not separately publish liquefaction fuel, which is embedded in pipeline and distribution use. Depending on future changes to the consumption series used elsewhere in the index, this proxy could introduce error. The Freeport check described above found no evidence of current distortion, but this should be reverified periodically.

Standing note for publication.

The index measures physical system conditions and does not evaluate whether any particular use of natural gas is socially or economically preferable. Growth in LNG exports will, other things equal, reduce this component. That is the measurement working as designed.

Sources. Natural Gas Monthly for production, consumption, and pipeline trade. LNG export series as a feedgas proxy, per the limitation above.

4.3 Global oil spare capacity cushion, 16.7 percent of total

Measures production that could be restored quickly if a major supplier goes offline.

Calculation. OPEC surplus crude production capacity divided by world liquid fuels consumption. Apply the Section 5.3 threshold function.

Definition, current as of December 2025. Effective production capacity is the rate reachable within 90 days and sustained without damaging fields, equal to maximum sustainable capacity minus disruptions. Surplus capacity is effective capacity minus actual production.

Definitional note. EIA defines surplus capacity as volume withheld under a coordinated OPEC or OPEC Plus agreement, excluding voluntary cutbacks from its definition of disruptions. This component therefore measures policy slack, not engineering headroom. This is appropriate for a security index, since withheld volume can be restored, but readers should not infer otherwise.

Interpreting a near zero reading. This component can legitimately score at or near zero while global oil production capacity remains substantial, and readers should understand why. Surplus capacity is effective capacity minus actual production. When OPEC Plus unwinds agreed production cuts, as it has been doing on a published schedule, actual production rises toward capacity and the unused margin closes. The physical ability to produce has not disappeared. What has disappeared is production that is available to be brought online quickly in response to an unplanned disruption elsewhere. That is precisely the buffer this index exists to measure, so a near zero score is the correct reading of a real condition, not an artifact. It does not mean the world is unable to produce oil.

Estimate rather than observation. OPEC spare capacity has no underlying historical statistical publication. Unlike inventories or refinery runs, it is not measured and then reported. EIA derives it from its own assessment of effective capacity against observed production, and publishes it through the Short Term Energy Outlook, which is a forecasting product. Every value in this component is therefore an EIA estimate at the vintage published, including values for months already past. This does not make the component unusable, since no better public measure of global spare capacity exists, but readers should understand that it carries a different evidentiary status from the inventory and storage components, and the page discloses this alongside the score.

Because this component can sit near zero for extended periods, the published page carries a contextual note alongside it rather than presenting the score without explanation.

Source. EIA Short Term Energy Outlook, monthly.

Backtest confirmed. This ladder used the full historical range without adjustment. The 2022 tightness period and the 2020 demand shock spike both scored sensibly, and this component is used as the reference example for how a well calibrated ladder should behave.

4.4 Oil inventory cover, 16.7 percent of total

Calculation. Three sub-metrics, scored individually against Section 5.4 thresholds, then weighted.

Sub-metric	Weight
US commercial crude days of supply	40%
SPR days of cover	30%
OECD commercial days of forward cover	30%

SPR days of cover measures strategic stock depth, not maximum emergency delivery rate. Drawdown and distribution capability are genuine constraints on how quickly reserve barrels can reach the market, and an eventual deliverability indicator is worth considering in a future version, but it does not block this one.

Backtest finding, SPR. The SPR score sat at or near the ceiling continuously from 2009 through early 2022, reflecting stable reserve policy over that period rather than a ladder defect. The metric only began discriminating meaningfully after the 2022 drawdown reset the baseline. This is expected behavior for a component tracking a policy stock rather than a market clearing quantity.

Backtest finding, crude days. During the April to June 2020 demand collapse, crude days of supply spiked to its highest score in the 2010 through 2025 record, driven by a collapse in refinery throughput rather than any improvement in physical security. This is a disclosed scope limitation rather than a corrected defect, consistent with the treatment of refinery utilization in Section 4.5. During acute demand collapses, this measure can read misleadingly comfortable because the throughput denominator falls faster than the headline stress condition.

Source. EIA Weekly Petroleum Status Report for crude and SPR stocks and refinery inputs. EIA Short Term Energy Outlook for OECD stocks and consumption.

4.5 Refined product cover, 16.7 percent of total

Calculation. Two sub-metrics, equally weighted, scored against Section 5.5 thresholds. Distillate days of supply and gasoline days of supply.

Naming. An earlier draft called this refining headroom, which did not match what it measures. The component is built entirely from distillate and gasoline days of supply, which are product inventory measures, not measures of spare refining capacity. Renaming it refined product cover makes the name match the measurement. Refinery utilization is deliberately not reintroduced to close the gap, for the reason given below.

Refinery utilization is excluded from the score. An earlier draft scored inverted utilization, which produced a genuine defect. When hurricanes or a demand collapse remove capacity from service, utilization falls, and the inverted score would have reported improved cushion. Unused capacity is not necessarily available capacity, since operable capacity includes idle and repair capacity. An asymmetric scoring function could correct the direction but would require defending an arbitrary threshold. Removing the input is cleaner, since the two product inventory measures already capture whether refining is delivering usable fuel. Utilization is retained on the published page as a diagnostic readout, not as a scored input.

Backtest finding, gasoline days. Gasoline days of supply shows the same 2020 pattern as crude days, spiking to its highest score in the record as product supplied collapsed faster than stocks drew down. The same disclosed scope limitation applies.

Backtest finding, distillate days. The late 2022 distillate tightness widely reported in the market registered correctly as a low score under this ladder. No adjustment needed.

Source. EIA Weekly Petroleum Status Report.

5. Threshold anchors

All anchors below have been backtested against 2010 through 2025 monthly data and are considered settled for this version, pending the ongoing monitoring items noted in Sections 4.2 and 4.4.

5.1 Gas storage adequacy

Demand weighted regional deviation from same week five year average, in percentage points.

Deviation	Score
-45 or worse	0
-30	20
-15	40
0	50

15	70
30	90
45 or better	100

5.2 Domestically retained supply

See Section 4.2 for the full table and rationale. Anchored on committed share from 7 percent, scoring 100, to 35 percent, scoring 0.

5.3 Spare capacity

Surplus capacity as a share of world liquids demand.

Share	Score
0%	0
1%	25
2.5%	50
4%	75
6% or above	100

The placement of 50 at 2.5 percent of world demand is the most consequential single number in the index and has been argued directly rather than fitted to the observed range.

5.4 Oil inventory cover

Crude days of supply

Days	Score
18 or fewer	0
22	25
27	50
32	75
38 or more	100

SPR days of cover

Anchors reflect the post 2022 reserve level, following the roughly 40 percent drawdown in absolute SPR stocks. Anchors calibrated to the pre 2022 world would read every recent year as permanently strained relative to a policy level that no longer applies.

Days	Score
12 or fewer	0
18	25
24	50
30	75
36 or more	100

OECD forward cover

Anchors are set from the observed distribution of this measure, which has run between 57 and 65 days across 2009 to 2026, with 2020 the sole outlier at 74. Fifty is placed at 60 days, the long run central value. An earlier version anchored 50 at 90 days on the basis of the IEA minimum stockholding obligation. That was an error: the IEA obligation is measured against net imports, whereas this measure is days against total consumption, and the two denominators differ by roughly half. The mistaken anchor placed the

entire historical range below the bottom of the ladder, so the sub-metric scored near zero throughout for reasons unconnected to actual conditions.

Days	Score
52 or fewer	0
56	25
60	50
65	75
72 or more	100

5.5 Refined product cover

Distillate days of supply

Days	Score
20 or fewer	0
25	25
32	50
38	75
45 or more	100

Gasoline days of supply

Days	Score
22 or fewer	0
24	25
27	50
30	75
33 or more	100

6. Display bands

Because 50 is defined as physically adequate, bands are substantive claims rather than distributional quantiles.

Band	Score range
Strained	0 to less than 25
Tight	25 to less than 40
Adequate	40 to less than 60
Comfortable	60 to 100

Intervals are closed at the lower bound and open at the upper bound for each band except the top, which is closed on both ends. Bands are set from the threshold definitions, not derived from backtest quantiles. Deriving them from the historical distribution would reintroduce relativity at the display layer immediately after removing it from the scoring layer.

7. Operating rules

Frozen thresholds and calibration. All anchors and band boundaries are fixed for this version and revised only at scheduled methodological releases, labeled 1.3 and onward. Revising a threshold in response to an inconvenient reading would be disqualifying.

No restatement. Published values are never revised. Source revisions enter the current week.

This rule has a recent worked example. EIA's December 2025 capacity update raised OPEC capacity estimates by 0.22 million barrels per day for 2024, 0.37 for 2025, and 0.31 for 2026, with similar increases to surplus capacity. A restating index would have silently rewritten two years of history for a component carrying 16.7 percent weight.

Backcast versus as published. Two clearly labeled regimes. From 2010 to launch, the index is a reconstructed historical series. From launch onward, it is an as published vintage series. Historical observations were not generated contemporaneously and the document does not imply otherwise.

Monthly inputs. Spare capacity and OECD cover update monthly, producing a step function in roughly a quarter of index weight.

Interpolation is prohibited. Interpolating between monthly observations requires the subsequent value, which is unavailable at publication. The alternatives are revising published figures, which violates the no restatement rule, or extrapolating, which presents constructed values as measurement. The step is disclosed in the weekly narrative instead.

Stale data. Hold the last value, flag it visibly on the page, and never drop a component and silently reweight the remainder.

Structurally lagging sources. The stale data rule above governs outages, meaning a release that was expected and did not arrive. It does not govern sources that publish on a permanently later schedule than others. The gas balance series drawn from the Natural Gas Monthly run roughly three months behind the weekly petroleum series, and this is a fixed feature of the source rather than a failure. Applying the outage rule to a permanent condition would leave the index flagged stale in nearly every period, which would train readers to disregard the flag when a genuine outage occurs. Components drawn from structurally lagging sources are therefore carried forward and marked at the component level, identifying the specific input and the number of months carried, rather than flagging the index as a whole. This treatment is permitted only where the affected component is slow moving enough that carry forward does not materially distort it. Domestically retained supply qualifies, since it is a trailing twelve month structural measure and repeating a small number of months inside a twelve month window changes it very little. Gas storage adequacy would not qualify, and must never be carried forward in this way.

Two source groups qualify under this rule. The gas balance series from the Natural Gas Monthly run roughly three months behind, affecting domestically retained supply. The Short Term Energy Outlook series run two months behind once estimates are excluded, affecting global spare capacity and the OECD part of oil inventory cover. Both are slow moving, so carrying a small number of months forward does not materially distort either, and each is marked at the component level on the published page.

Closed months only. The weekly petroleum series accumulate through a month, so a figure computed for a month still in progress changes every time the index is rebuilt. Publishing such a figure as the headline would mean revising an already published value, which the no restatement rule forbids. The headline therefore reports the last completed month. A figure for the month in progress is shown separately, labelled as provisional, and is explicitly not a published value. Only completed months are written to the archive, and an archived month is never rewritten.

Estimates are not observations. The Short Term Energy Outlook publishes three kinds of row: historical, estimate, and forecast. EIA distinguishes them by shading in its published tables but not in the data interface, so no programmatic marker is available. Excluding rows dated in the future removes forecasts but leaves estimates, which are not observations. Series drawn from the Short Term Energy Outlook are therefore cut two months back from the current month, which approximates the boundary given that

modelling for each release is completed at the start of its release month. This is an approximation and is disclosed as one.

Reproducibility. Every input is public, every transformation is published, and an outside analyst with a spreadsheet can rebuild the index. This is treated as the index's principal long term asset.

8. Validation

8.1 Correlation matrix

Computed across all five scored components on 156 months of complete data, including the full crude, SPR, and OECD weighting for oil inventory cover.

	Spare capacity	Inventory cover	Refining	Storage adequacy	Uncommitted share
Spare capacity	1.00	0.18	0.40	0.16	-0.31
Inventory cover	0.18	1.00	0.70	0.21	0.24
Refining	0.40	0.70	1.00	0.42	0.08
Storage adequacy	0.16	0.21	0.42	1.00	-0.24
Uncommitted share	-0.31	0.24	0.08	-0.24	1.00

Inventory cover and refining correlate at 0.70. This is expected, since both draw on overlapping crude and product stock data during active drawdowns.

Refining and gas storage adequacy correlate at 0.42, higher than the near independence originally assumed in an earlier draft. A shared winter weather driver was the first hypothesis, but a seasonal split of the data does not support it. Winter months, November through March, show a correlation of 0.34, while summer months, May through September, show a higher correlation of 0.48. This rules out winter weather as the explanation. A shared demand driver operating year round, or a structural link between summer gas injection season and peak driving season refinery runs, are both plausible alternatives, but neither has been confirmed. This relationship is disclosed here with the cause left open rather than asserted, and is flagged for further investigation in a future version.

Domestically retained supply correlates negatively with spare capacity, at negative 0.31, and with storage adequacy, at negative 0.24. Both are plausible and not treated as concerning, though the uncommitted supply share ladder carries the monitoring note described in Section 4.2.

8.2 Weighting sensitivity

The published 50 to 50 sub-index structure was compared against a flat 20 percent weighting across all five components, using the same 156 months of complete data.

Statistic	Result
Correlation between the two schemes	0.9846
Mean absolute difference	1.47 points
Maximum absolute difference	3.94 points, February 2021
Months where the display band changed	10 of 156, 6.4%

Weighting choice is largely immaterial to the index's output. The two schemes disagree on display band in roughly one month in sixteen, and every disagreement occurs near a band boundary rather than

reflecting a large substantive difference. This finding supports the choice of the simpler, more defensible 50 to 50 structure over a scheme requiring additional justification.

8.3 Episode validation

Individual components were checked against known historical stress episodes during the threshold review process described in Sections 4.1 through 4.5. Late 2022 distillate tightness, the 2014 and 2021 gas storage stress periods, and the 2022 OPEC tightness period all registered correctly. The April to June 2020 demand collapse produced known, disclosed false readings in crude days and gasoline days of supply, addressed in Sections 4.4 and 4.5 rather than corrected through ladder adjustment.

9. Companion indicators

Displayed on the page, explicitly outside the index.

Curve structure. The front month minus the fourth month WTI contract, divided by the front month, twenty day average. Excluded because it is a price input and the index's central claim is that it is not a price index. Requires a licensed data source, since EIA's free daily futures series ends in April 2024 and has not resumed. Licensing is resolved separately and does not block launch.

Replacement momentum. New well production divided by the absolute value of existing production change, from the Short Term Energy Outlook table that absorbed the discontinued Drilling Productivity Report, supplemented by the Baker Hughes rig count release. Excluded on data quality grounds and because it is a flow measure among otherwise stock measures. Retained because it carries the investment and permitting discussion.

Refinery utilization. Diagnostic readout only, per Section 4.5.

10. Publication

Schedule. Thursdays, following the release of the Weekly Natural Gas Storage Report, the last major weekly input. A fixed cadence builds readership habit, and habit produces citation.

Page contents. Headline gauge. Oil and gas sub-index gauges. A twenty four month trend line with the adequate threshold marked. Component bars grouped by sub-index and sorted from worst to best. A short narrative describing what moved the index that week. A band legend with threshold values. Companion charts. A CSV download. This methodology document. The embeddable version is the headline card alone.

Weekly narrative. Generated from the same underlying data as the index itself. Identifies the largest weighted change, the second largest, and one notable exception to the broader trend. Reviewed by a person before publication for at least the first six months.

Architecture. A scheduled job pulls machine readable EIA data and the Baker Hughes rig count release, writes a single data file, and a static page renders it. No database and no live backend are required.

11. Revision history

Version 1.6. Corrected the OECD forward cover ladder, which had been anchored to the IEA net imports stockholding obligation and applied to a days against consumption measure, placing the whole historical range below the bottom of the ladder. Anchors are now set from the observed distribution. Extended the

structurally lagging source rule to the Short Term Energy Outlook series, so global spare capacity and OECD forward cover are carried forward and marked rather than holding the headline two months back.

Version 1.5. Added the closed month rule, so the headline reports the last completed month and a month in progress is shown separately as provisional. Added handling for Short Term Energy Outlook estimates, which are excluded by a two month cutoff rather than treated as observations. Renamed uncommitted supply share to domestically retained supply, and refining headroom to refined product cover, so both names match what they measure. Disclosed that OPEC spare capacity is an EIA estimate at every vintage rather than an observed statistic. Corrected the gas storage baseline to key observations by ISO year rather than calendar year, which had misassigned the turn of year weeks.

Version 1.4. Rebuilt the gas storage region weights from state level January residential plus commercial consumption, replacing development placeholders, and corrected the state to region mapping against EIA's published definitions. Added an explanation in Section 4.3 of how to interpret a near zero spare capacity reading, following EIA data showing OPEC surplus capacity approaching zero as OPEC Plus unwound agreed production cuts. Added a Section 7 rule distinguishing structurally lagging sources from outages, permitting component level carry forward where the affected component is slow moving.

Version 1.3. Widened the domestically retained supply committed floor from 25 percent to 35 percent after review found the original floor would saturate within 4 to 5 years given the current pace of LNG buildout. Replaced the winter weather explanation for the refining to gas storage correlation with an open disclosure after a seasonal split of the data, winter 0.34 versus summer 0.48, ruled out the original hypothesis.

Version 1.2. Widened the gas storage ladder after backtest review found saturation at the original anchors. Added OECD forward cover as the third component of oil inventory cover, at full 40, 30, 30 weighting. Replaced the domestic gas supply margin with domestically retained supply after the margin was shown to be a near accounting identity correlated with storage adequacy at 0.62. Reviewed and confirmed the domestically retained supply ladder against real historical range and event checks. Added disclosed scope limitations for crude and gasoline days of supply during acute demand collapses. Added a note on the SPR score plateau from 2009 through early 2022. Ran the correlation matrix and weighting sensitivity test on complete, final data.

Version 1.1. Replaced percentile scoring with fixed thresholds. Set display bands from the thresholds rather than backtest quantiles. Added the dynamic range diagnostic and threshold sensitivity testing. Added tiered anchor disclosure.

Version 1.0. Adopted the 50 to 50 sub-index structure. Added the domestic gas supply margin, later replaced in version 1.2. Moved regional storage weighting into launch scope. Removed refinery utilization from scoring. Updated the spare capacity definition to the December 2025 EIA revision. Froze the calibration window. Adopted four week average denominators.

Appendix. State to storage region mapping

Used to build the regional weights described in Section 4.1. Verified against EIA's published storage region definitions. EIA last redefined these regions in October 2015. Alaska and Hawaii are excluded because the weekly storage series cover the Lower 48 only.

Region	Weight	States
East	0.4052	Connecticut, Delaware, District of Columbia, Florida, Georgia, Maine, Maryland, Massachusetts, New Hampshire, New Jersey, New York, North Carolina, Ohio, Pennsylvania, Rhode Island, South Carolina, Vermont, Virginia, West Virginia
Midwest	0.3008	Illinois, Indiana, Iowa, Kentucky, Michigan, Minnesota, Missouri, Tennessee, Wisconsin
South Central	0.1118	Alabama, Arkansas, Kansas, Louisiana, Mississippi, Oklahoma, Texas
Mountain	0.0924	Arizona, Colorado, Idaho, Montana, Nebraska, Nevada, New Mexico, North Dakota, South Dakota, Utah, Wyoming
Pacific	0.0898	California, Oregon, Washington